

Audio LF-CBM Concept Construction Pipeline

1 Task Inputs

Example Datasets



ESC-50

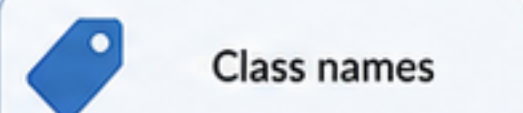


UrbanSound8K

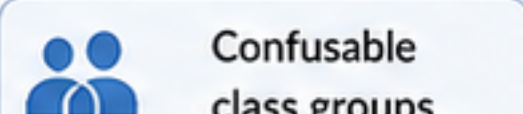


CREMA-D

Prompt Sources



Class names



Confusable class groups

Audio domain only



2 Prompting Strategies



Vanilla LF prompts

Generate audible characteristics for each class.

- e.g., pitch, timbre, rhythm, texture, energy, attack-decay



Broad prompts

Generate short atomic auditory properties; forbid class names, object identities, emotions, visual descriptions.

- e.g., high pitched, rough texture, low energy, slow decay



Contrastive prompts

Given acoustically confusable class groups, generate discriminative auditory dimensions.

- e.g., pitch instability, modulation rate, spectral centroid shift

3 Raw Concept Candidates

LF concepts

guitar strumming

jackhammer pounding

dog barking bursts

⋮

Broad concepts

high pitched

rough texture

slow decay

low energy

⋮

Contrastive concepts

pitch instability

rapid amplitude modulation

steady low-frequency hum

⋮



Full vocabulary =

LF $\cup$ Broad $\cup$ Contrastive

4 Filtering & Consolidation

Apply filters

- ✓ remove class-name leakage
- ✓ deduplicate / merge near-synonyms
- ✓ keep short atomic phrases
- ✓ CLAP screening
- ✓ projection screening
- ✓ retain interpretable concepts

Output vocabularies



LF



LF + Broad



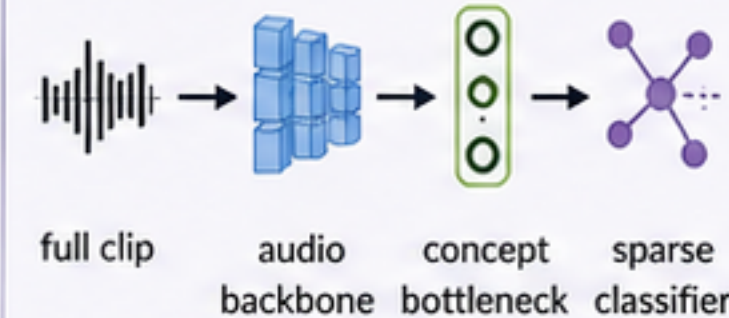
LF + Contrastive



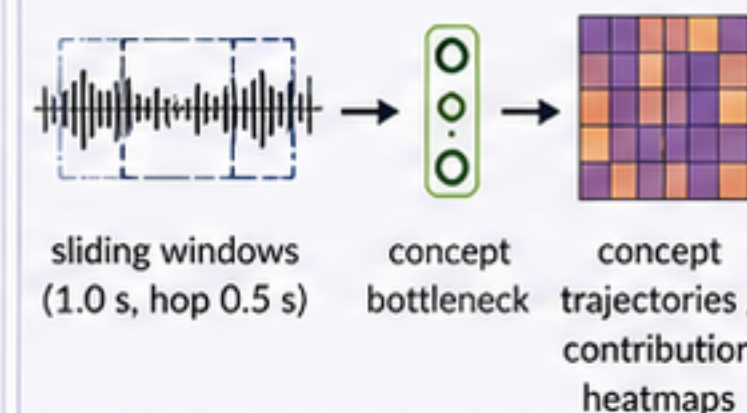
Full Union

5 Use in LF-CBM

Global Inference



Temporal Inference



💡 Temporal analysis reuses the same learned vocabulary; only inference is windowed.



Concept generation happens **once**; temporalization happens at **inference**.